Experiment:-8

Problem Statement:-

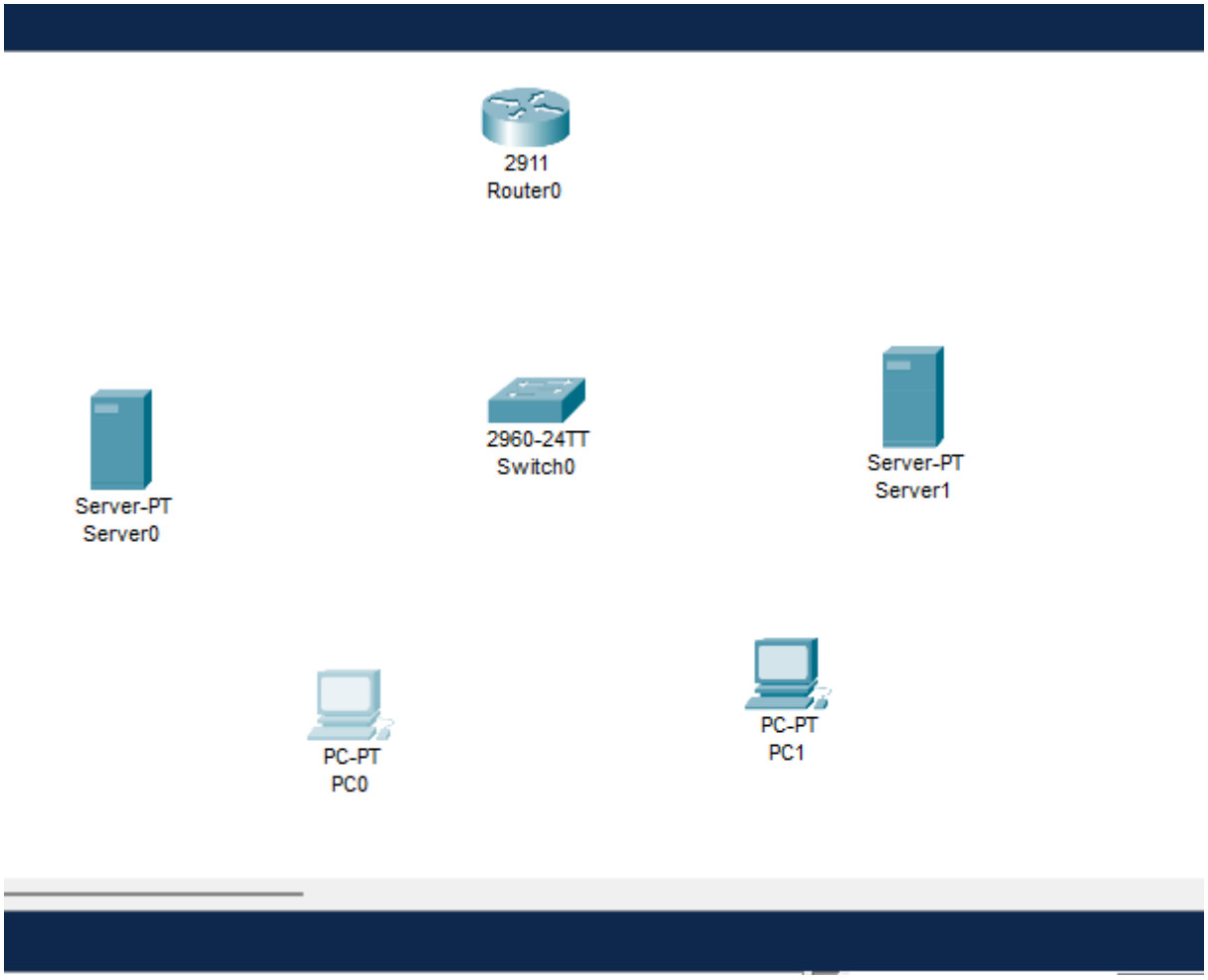
Create a simulation to demonstrate logical addressing using IPv4 and IPv6. Implement address mapping techniques such as ARP, RARP, BOOTP, and DHCP to show how devices acquire and resolve network addresses. Use Cisco Packet Tracer to visualize and analyze these address resolution processes.

Aim:- **Simulation of Logical Addressing using IPv4 and IPv6**

Procedure:-

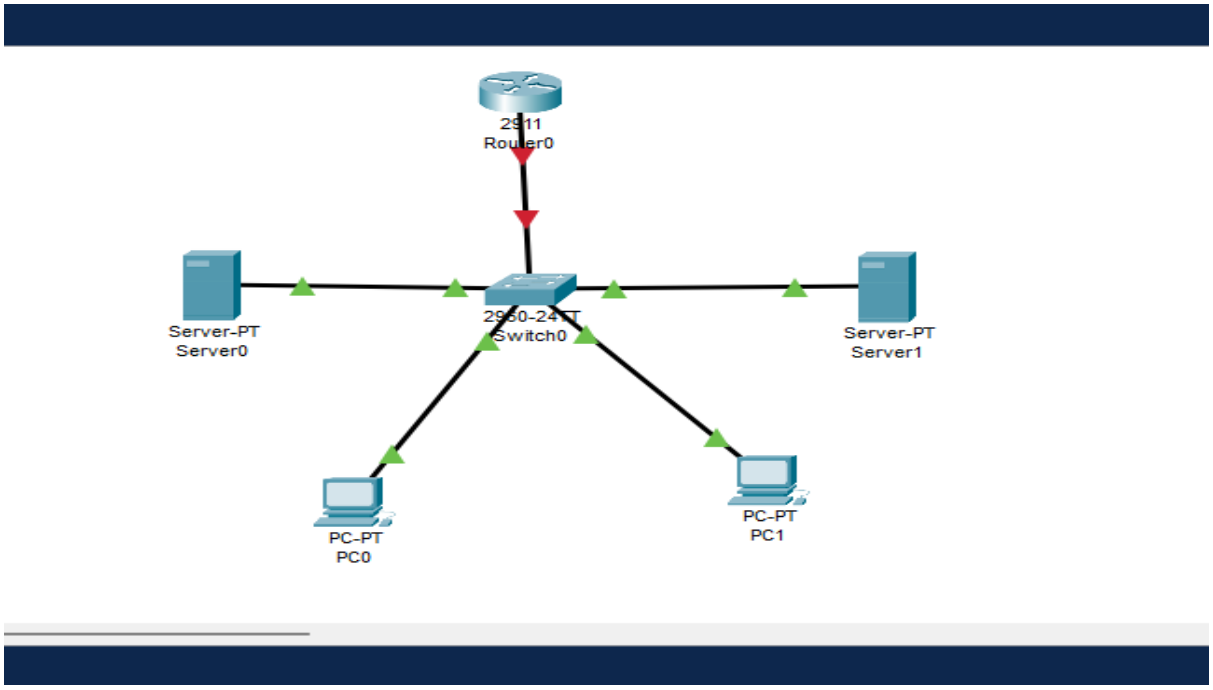
Step 1: Add All Devices

1. Open Cisco Packet Tracer
2. Add these devices to workspace:



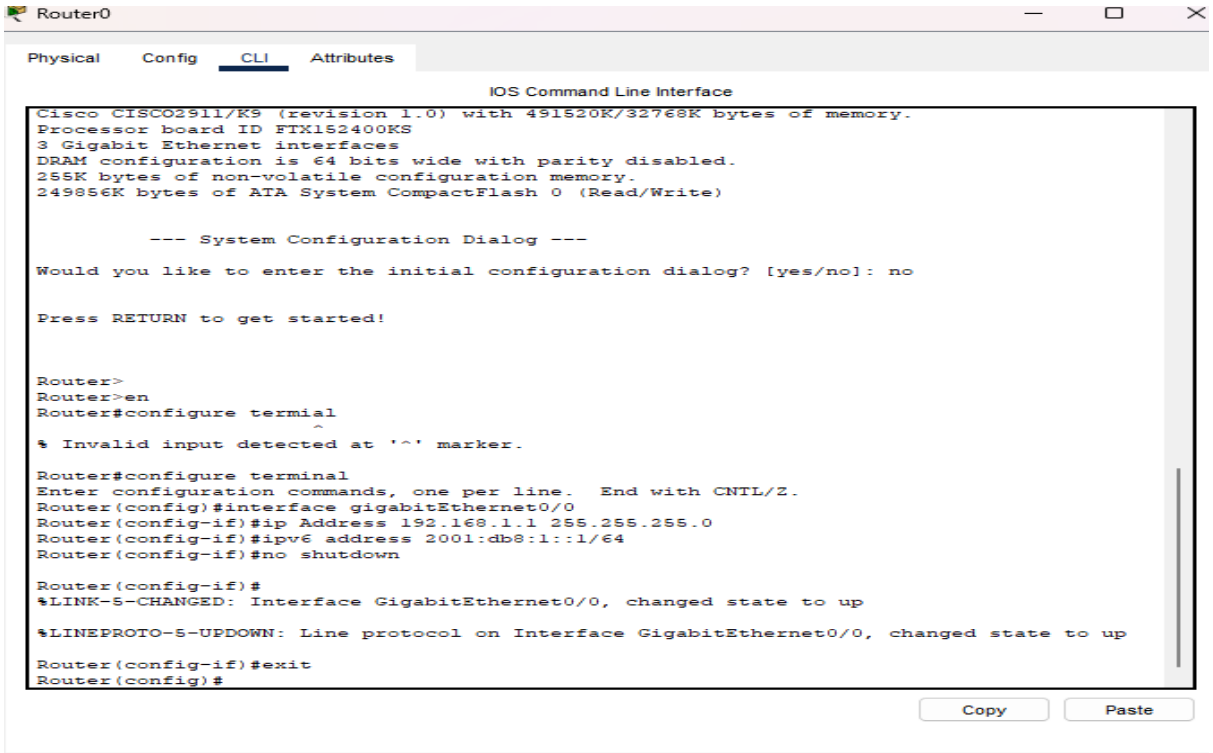
Step 2: Connect All Devices

Use **Straight-Through Copper Cable** for all connections:



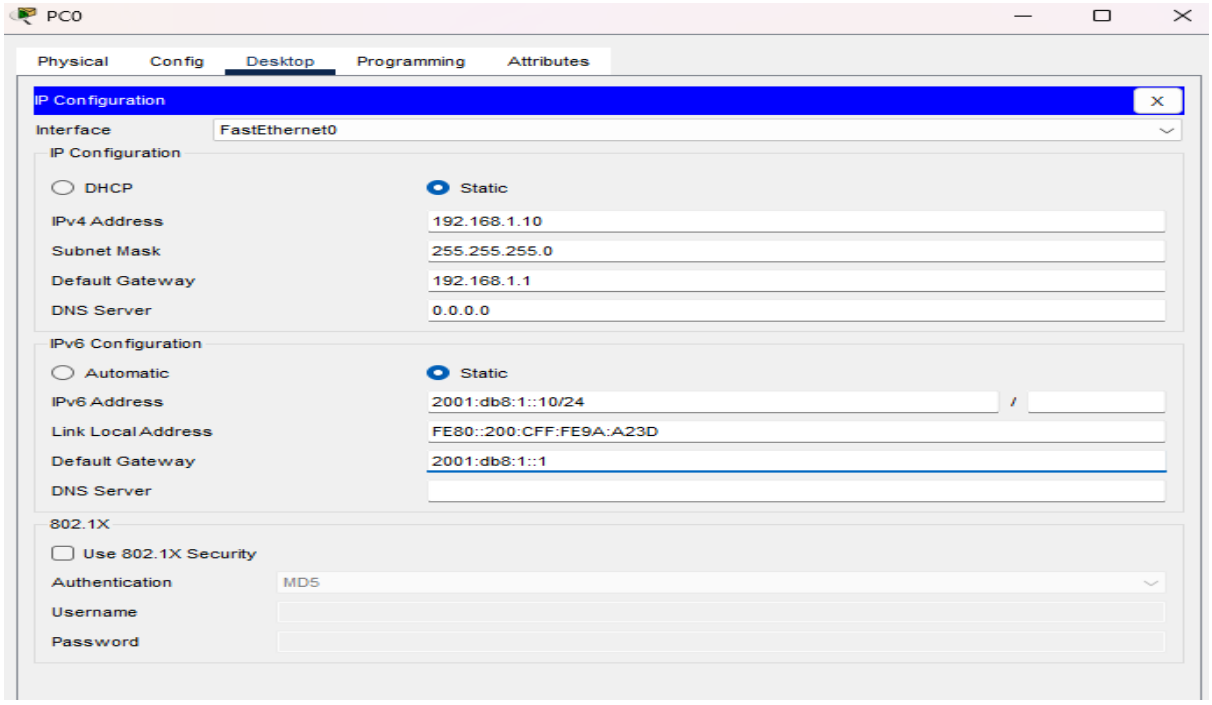
Step 3: Configure Router Interface

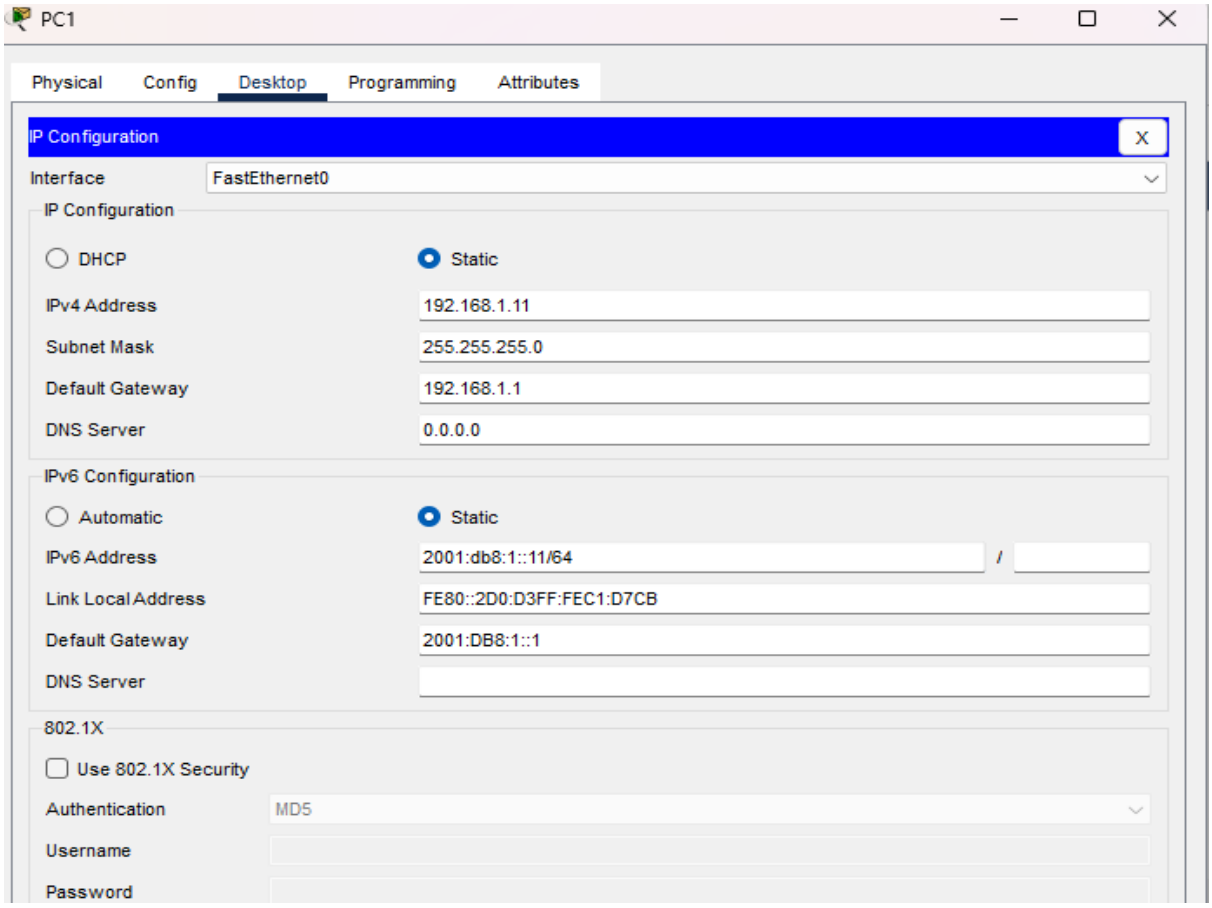
1. Double-click **Router**
2. Go to **CLI** tab
3. Type these commands:



Step 4: Assign Static IP to PC0

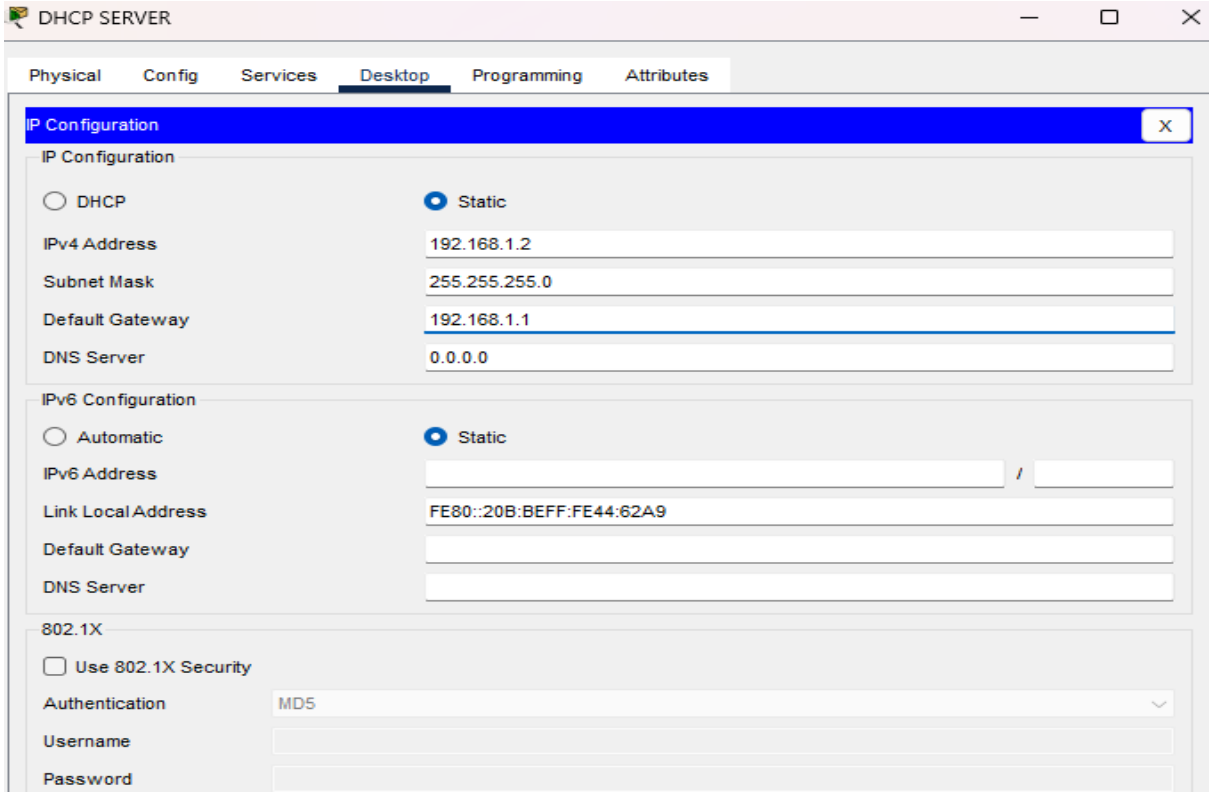
1. Double-click **PC0**
2. Go to **Desktop → IP Configuration**
3. Enter:

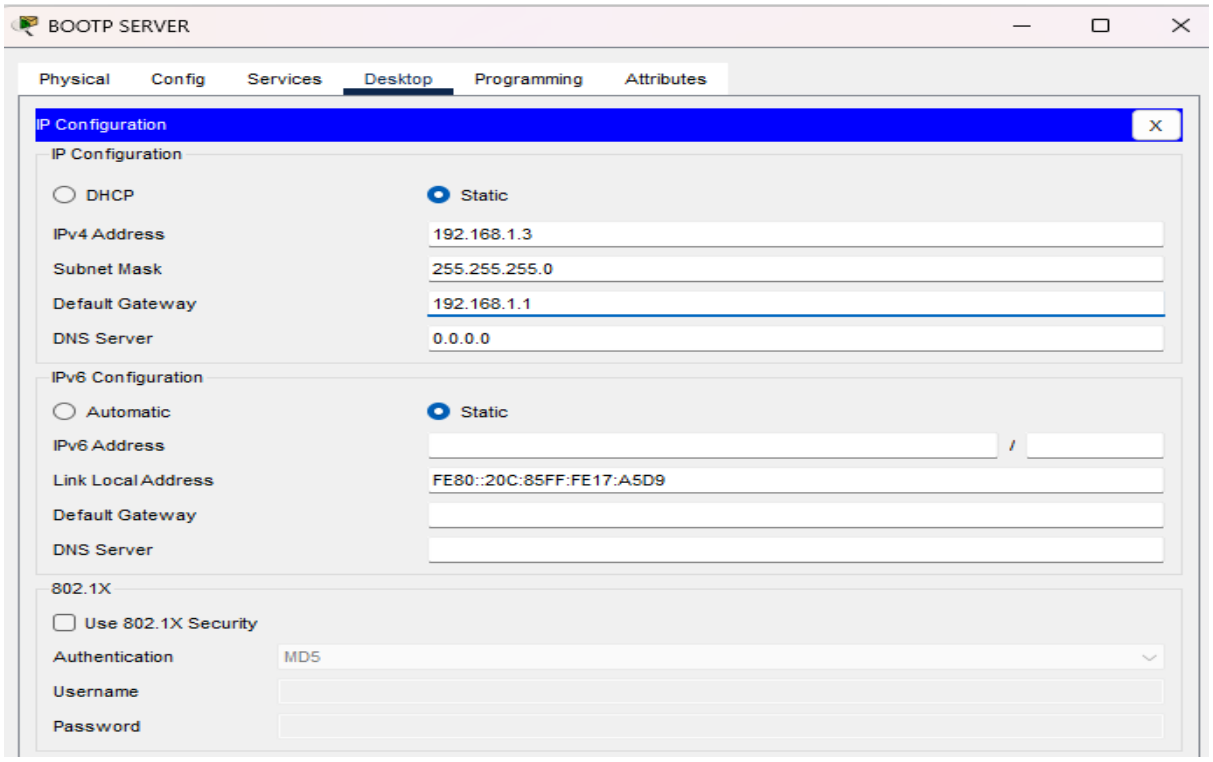




Step 6: Assign IP to DHCP Server

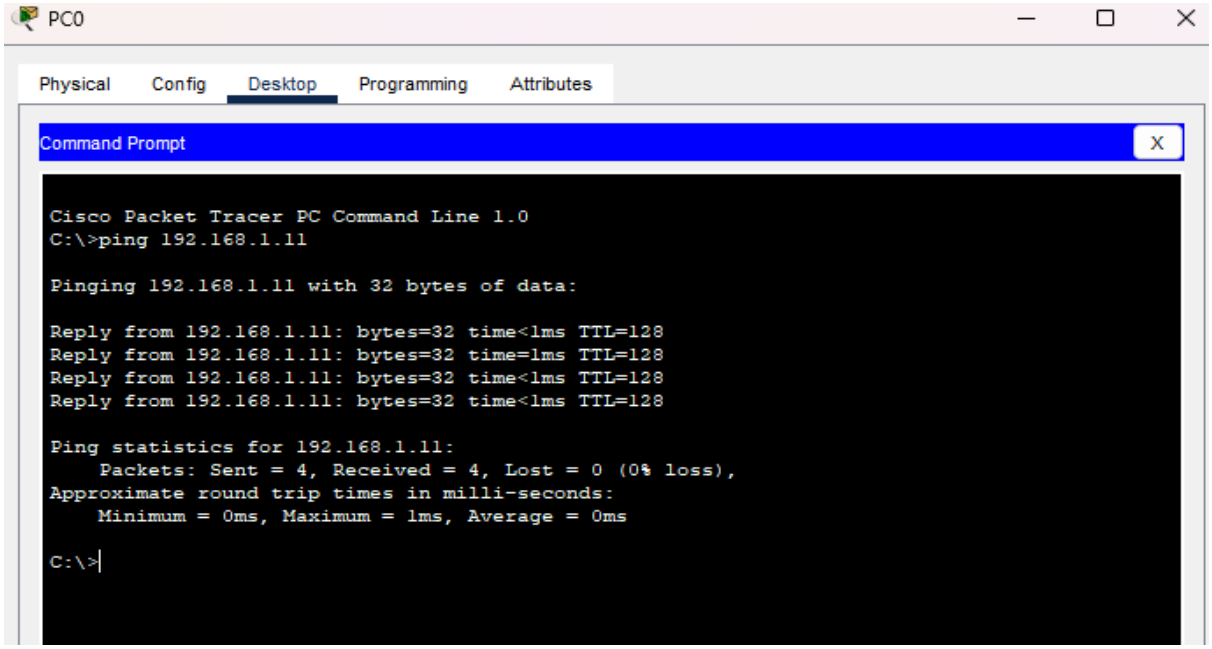
1. Double-click **DHCP Server**
2. Go to **Desktop → IP Configuration**
3. Enter:





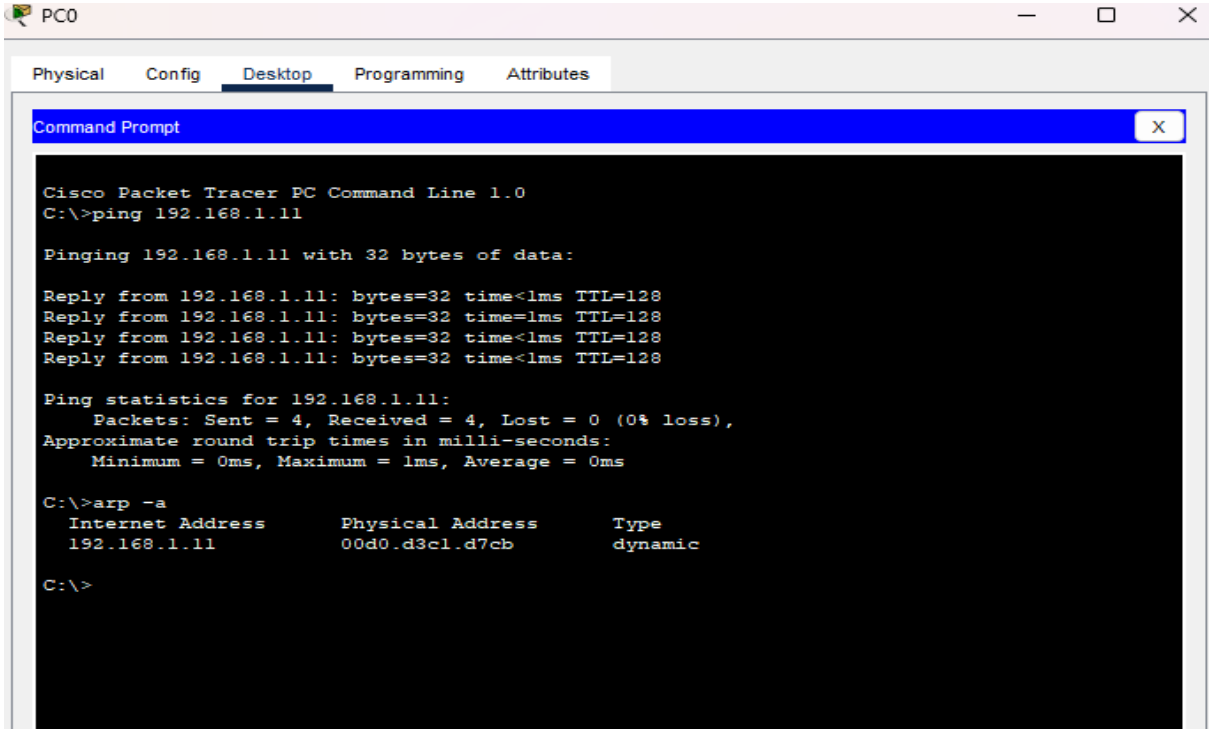
Step 8: Simulate ARP Request

1. Double-click **PC0**
2. Go to **Desktop → Command Prompt**
3. Type:



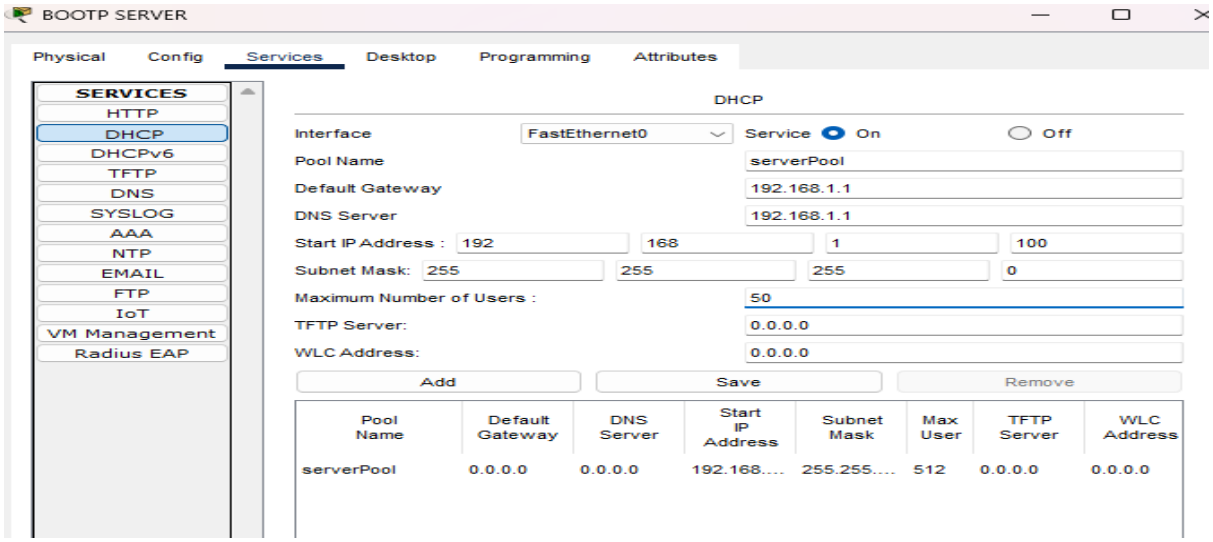
Step 9: View ARP Table

After ping, type in PC0 Command Prompt:



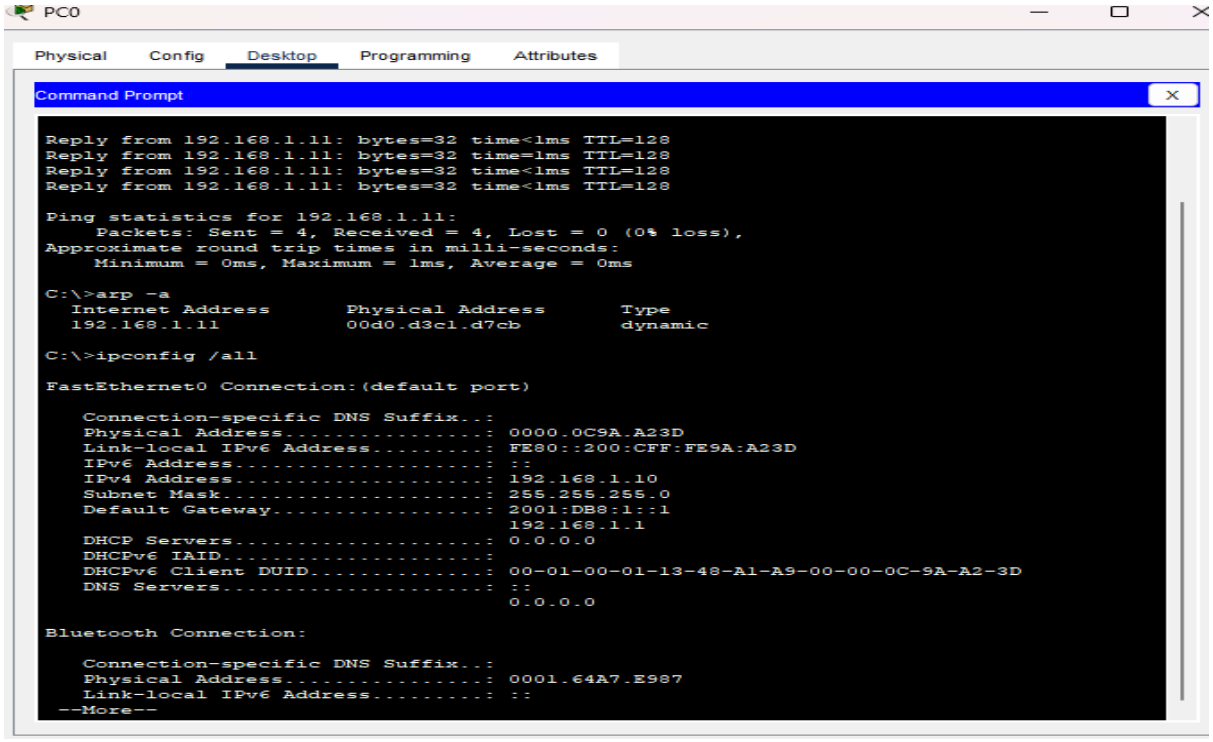
Step 10: Simulate RARP Concept

1. Double-click **BOOTP Server**
2. Go to **Services** tab
3. Click **DHCP** service
4. Add a static mapping entry:



Step 11: Find PC0's MAC Address

1. Double-click **PC0**
2. Go to **Desktop → Command Prompt**



RARP (Reverse ARP):

- Device sends its MAC address

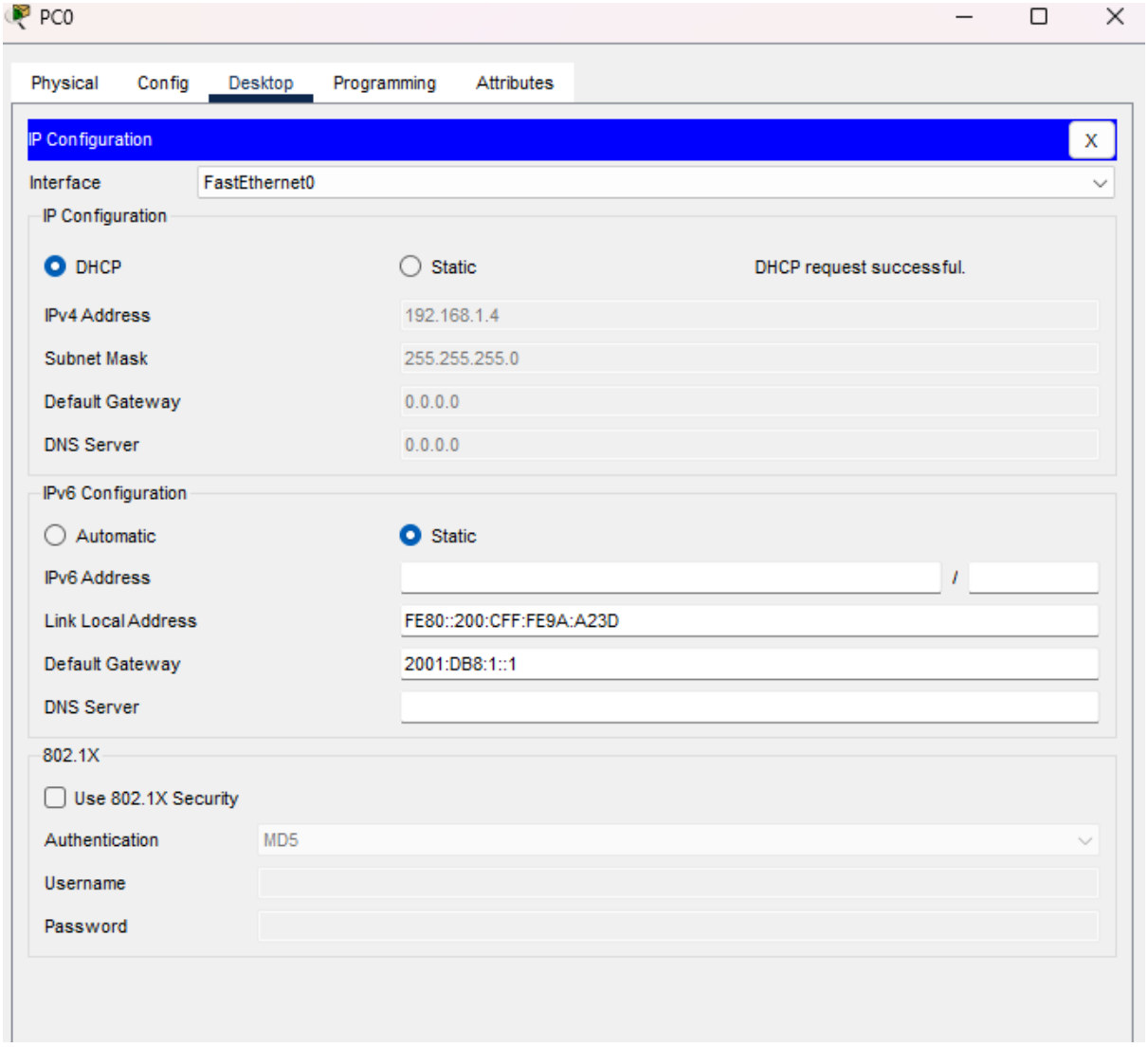
- Server responds with IP address

- Legacy protocol replaced by BOOTP/DHCP

- Simulated here using static DHCP mapping

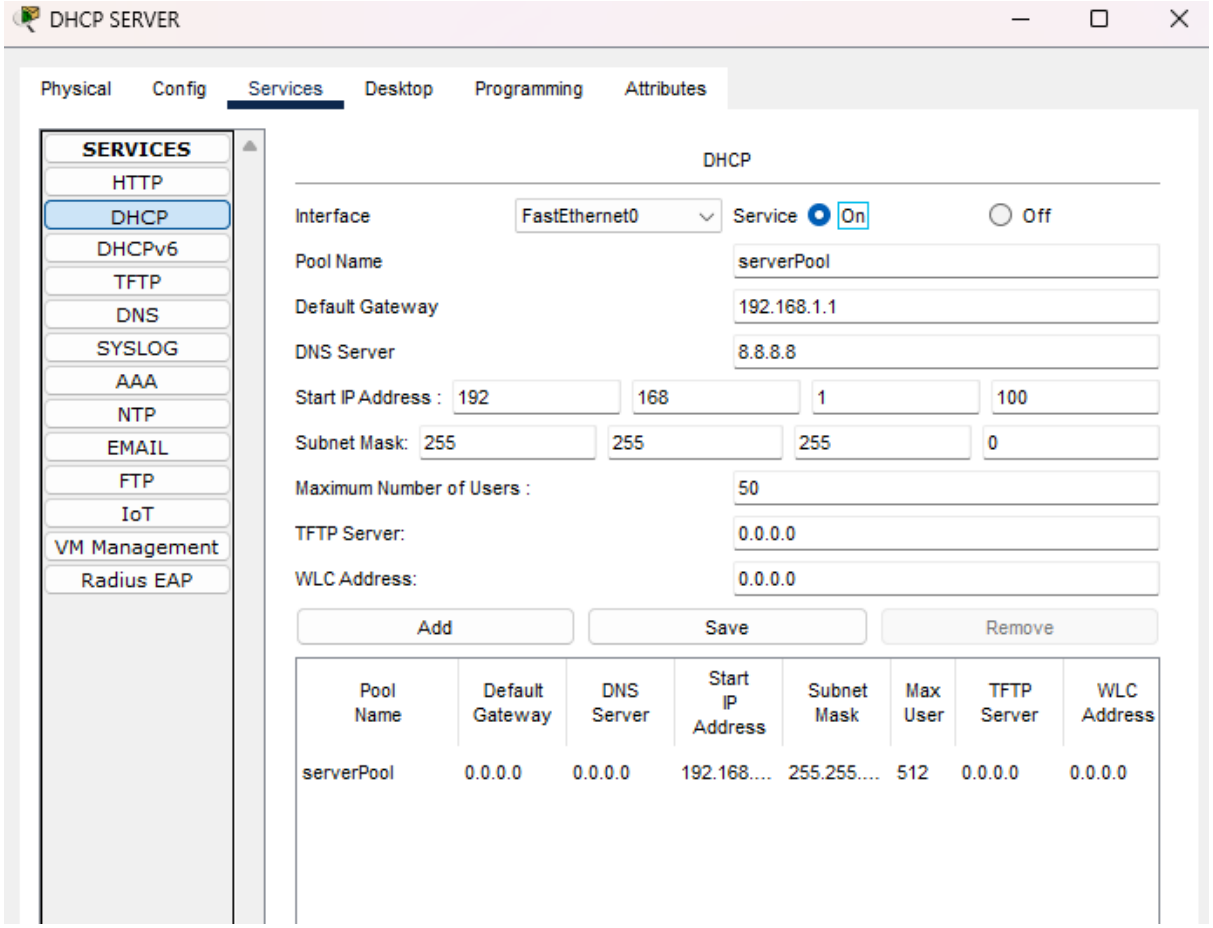
Step 13: Test BOOTP on PC0

1. Double-click **PC0**
2. Go to **Desktop → IP Configuration**
3. Select **DHCP** radio button
4. PC0 will send a **BOOTP/DHCP Discover** broadcast
5. Server responds with an IP address
6. You'll see IP auto-assigned



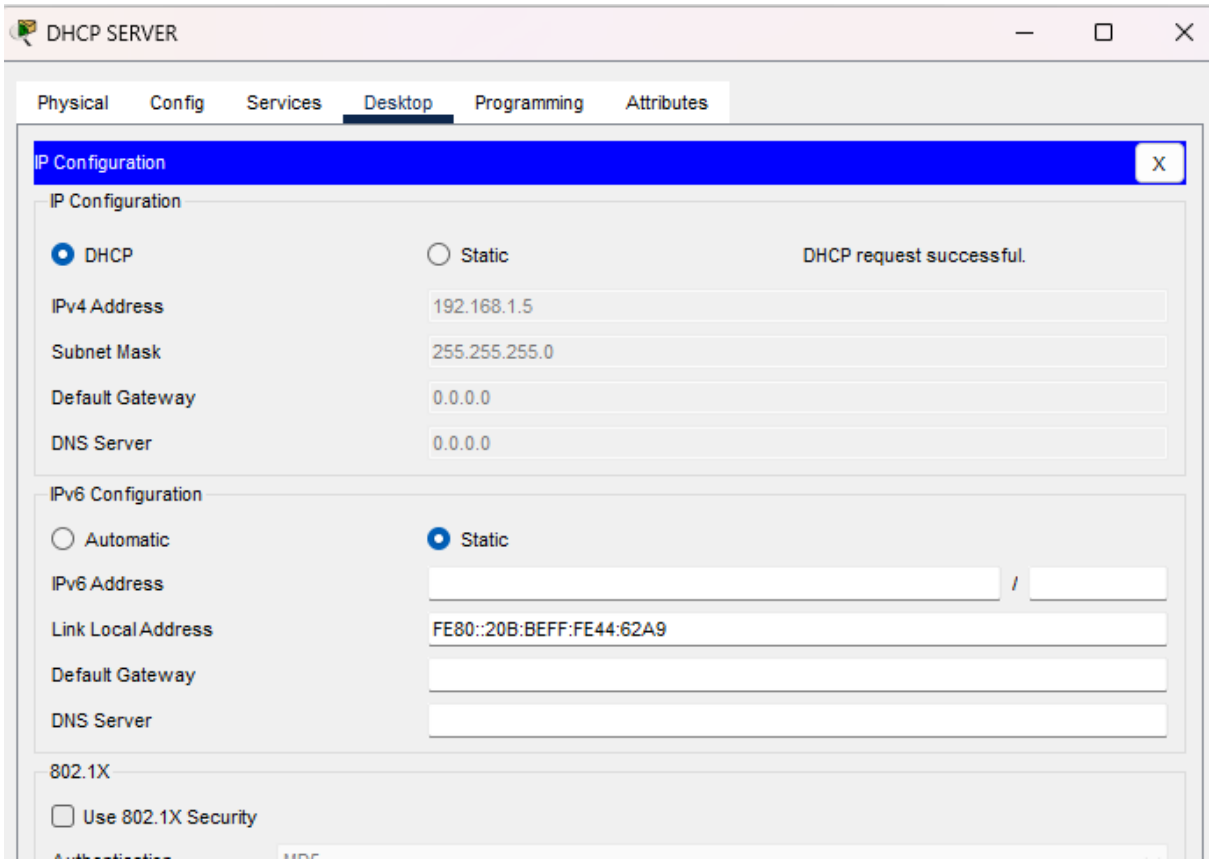
Step 14: Enable DHCP Service

1. Double-click **DHCP Server**
2. Go to **Services** tab
3. Click **DHCP**
4. Configure the pool:



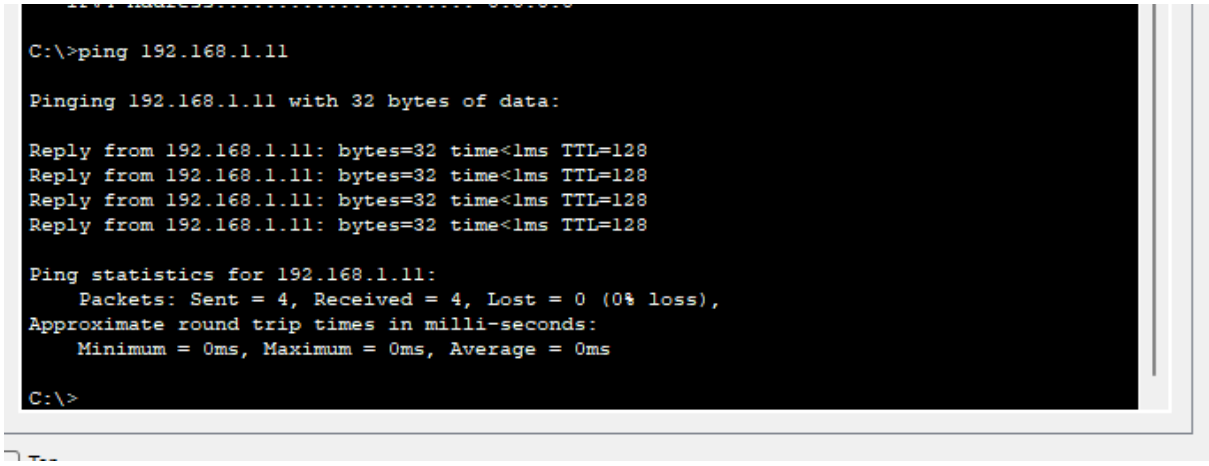
Step 15: Test DHCP on PC1

1. Double-click **PC1**
2. Go to **Desktop → IP Configuration**
3. Click **DHCP** radio button
4. Wait 5–10 seconds
5. PC1 will receive:



Step 16: IPv4 Ping Test

On PC0 Command Prompt:



Step 17: IPv6 Ping Test

On PC0 Command Prompt:

```
C:\>ping 2001:db8:1::1

Pinging 2001:db8:1::1 with 32 bytes of data:

Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Step 19: Ping the Router

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

```
C:\>ping 2001:db8:1::1

Pinging 2001:db8:1::1 with 32 bytes of data:

Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:1::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:1::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```